The Limits of Predictions for Online Bipartite Matching: A Unified Experimental Study and a Budget–Stakes Law

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Learning-augmented ("with-predictions") algorithms for online bipartite matching have proliferated: Min-PredictedDegree consumes a per-node degree predictor, and a family of *test-and-fallback* schemes tests a type-histogram prediction on a sublinear prefix of the arrivals before committing to follow it or to fall back on an advice-free baseline. These algorithms have been analyzed in isolation — on different input models, against different error models, and largely in theory. We give the first unified experimental study, placing all of them on a single harness with a common structured prediction-error model, a common optimum, and confidence intervals, across synthetic graphs, six real-world graphs, and real request traces. Our central finding is that, on average-case (known-i.i.d.) inputs, the value of predictions is **robustness insurance, not a performance lever**: the advice-free baseline is already near-optimal, so the *consistency* upside of good advice is small, whereas unguarded prediction-following can crash far below the baseline — and the practical worth of the sophisticated algorithms is precisely that they never do. We then prove what the wall *costs*. For the test-and-fallback class we establish a two-sided **budget–stakes law**: deciding whether to follow the advice requires — and, via an explicit one-line rule, only requires — a prefix of length $\tilde{\Theta}(\sigma^2/g^2)$: the instance's specialist mass σ^2 (exactly twice its baseline slack) divided by the square of the stakes g (what good advice gains over the baseline), independent of the instance length. The two halves meet, up to logarithms, whenever the stakes are carried at comparable signal levels by a constant fraction of that mass; one low-signal regime is left open and is stated as such. The lower half binds *every* decision rule; the upper half refutes the natural conjecture that the hardness of tolerant distribution testing blocks all sublinear rules — the decision-relevant statistic is the *payoff* of following, which is exponentially cheaper to test than the prediction's distance — a prescription we validate on the benchmark, where a constant-free payoff-testing rule avoids the threshold pathologies of the deployed tests. Because the stakes are capped by the baseline slack, on strong-baseline instances the price exceeds the horizon: upsides below $\approx \sqrt{(1-\rho_{\text{base}})/n}$ are uncapturable by any rule at any prefix length, and the upsides we measure sit in that range. Experiments and theory deliver one message: **on average-case matching, the advice's upside is smaller than the price of finding out whether to trust it.**

CCS Concepts: • **Theory of computation → Online algorithms**.

Additional Key Words and Phrases: online bipartite matching, algorithms with predictions, learning-augmented algorithms, consistency–robustness trade-off, distribution testing, experimental algorithmics

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1 Introduction

Online bipartite matching is the canonical model of irrevocable sequential allocation: offline resources are known in advance, requests arrive one at a time, and each request must be matched to an available neighbor — or dropped — immediately and irrevocably, before the rest of the input is seen. It underlies ad allocation, ride-hailing dispatch, organ exchange, and request routing in serving systems. Classical online algorithms — Greedy, KVV Ranking [16], and the stochastic-matching algorithms of Feldman et al. [11] and Jaillet–Lu [14] — are analyzed through the worst-case competitive ratio.

A now-large body of work augments these algorithms with a *prediction* about the input and asks for two guarantees at once: **consistency** (near-optimal performance when the prediction is good) and **robustness** (no worse than an advice-free baseline when the prediction is arbitrarily bad) [17, 21]. For online matching specifically, two strands have emerged. MinPredictedDegree (MPD) [1] matches each arrival to its available neighbor of minimum *predicted* degree, protecting scarce resources; it is robust by construction (a useless predictor reduces it to Ranking). A second strand — the *test-and-fallback* algorithms of Choo et al. [9] and Burathep–Erlebach–Moses [6] — is explicitly adaptive: it Mimics a type-histogram prediction on a sublinear prefix of arrivals, tests whether the observed arrivals match the prediction, and then either continues to follow it or falls back to Ranking.

These algorithms have been studied *in isolation* — each on its own input family, against its own error model, and largely through worst-case theory. As a result, three basic questions have no empirical answer. How do the algorithms actually compare, head to head, under one prediction-error model? How much does a prediction help, and at what cost, on realistic inputs? And — most importantly — *why* does the practical experience with these algorithms feel so different from their worst-case promise? This paper answers all three, with a unified experimental study and a theorem that explains what the study finds.

A unified experimental study. We place the advice-free baselines, the MPD family (and its Feldman/Jaillet–Lu augmentations), and the test-and-fallback algorithms on a single harness: common graph families, a common structured prediction-error model, a common optimum computed by Hopcroft–Karp, and 95% confidence intervals throughout. We also port the blind-follow-with-switching *combiner* of Chłędowski et al. [8] — the "cheap worst-case insurance" from the caching literature — and benchmark it (we do not claim it as new). The study runs across synthetic graphs spanning the difficulty range, six real-world graphs from the Network Repository, and real request traces.

The empirical wall. Across every setting we observe the same phenomenon, which we state as the paper's organizing thesis: *on average-case inputs the value of predictions is robustness insurance, not a performance lever.* Concretely (Section 3): the advice-free Ranking is already within a percent or two of the optimum, so the consistency upside of even perfect advice is small; unguarded prediction-following (blind Mimic, or MPD under adversarial predictions) crashes *below* the advice-free floor; and the entire practical value of the sophisticated algorithms is that they refuse to crash. Two distinct mechanisms deliver this — the structural robustness of the MPD-augmentations and the adaptive robustness of test-and-fallback — with different consistency/robustness trade-offs. We confirm the wall on the six real graphs and on real traces, where a cheap, non-ML historical predictor captures a meaningful fraction of the (small) oracle gap while never dropping below the baseline (Section 6). We also report a negative result: because MPD depends on its predictor only through the induced *order*, one might train the predictor with an order-aware loss instead of regression, but on realistic features the two coincide — reinforcing that the predictor, once order-faithful, is not the bottleneck.

Why the wall is necessary — and what it costs. The empirical wall is not an artifact of a particular generator or algorithm; it has a price tag. Our main theoretical contribution (Section 7) is a two-sided **budget–stakes law** for the test-and-fallback class. Informally:

> Deciding whether to follow the advice costs a prefix of length $k^* = \tilde{\Theta}(\sigma^2/g^2)$, where g is the stakes (what good advice gains over the baseline) and σ^2, the arrival mass on contested resources, equals exactly twice the baseline slack. Below k^*, *no* decision rule is simultaneously consistent and robust; at k^*, an explicit one-line rule is.

The two bounds meet up to logarithms whenever the stakes are carried, at comparable signal levels, by a constant fraction of the contested mass — the condition under which "k^*" names a single quantity; §7.8 records the low-signal regime in which the two sides can still separate, which we leave open. The lower half is information-theoretic — it binds *any* measurable rule on the prefix, not merely the empirical-distance threshold used in practice — via a master consistency/robustness inequality and a Hellinger computation. The upper half is a **directional test**: classify each prefix arrival as agreeing or disagreeing with the advice's local prediction, and follow iff agreements win — a rule we also implement, calibrate, and evaluate head-to-head on the benchmark (§5.4). Its analysis rests on a *payoff identity* — on the hard family, the advantage of following is exactly half the mean of a per-sample observable — which also refutes the tempting stronger conjecture (and an earlier version of our own claim) that tolerant-testing hardness [7] makes the decision impossible for every sublinear rule: the advice's *distance* to the truth is indeed hard to test, and the field's empirical-ℓ_1 thresholds are provably blind at sublinear prefixes, but the decision-relevant *payoff* is not. The wall then re-emerges exactly where the experiments live: stakes are capped by the baseline slack ($g \leq 2\varepsilon_{\max}(1-\rho_{\text{base}})$, with $\varepsilon_{\max}$ the largest per-cell signal), so on strong-baseline instances the budget k^* exceeds the horizon itself — every upside below $\Theta(\sqrt{(1-\rho_{\text{base}})/n})$ is uncapturable at *any* feasible prefix, and the upsides measured in Sections 3–6 sit in that range.

Contributions. - **(C1) The first unified benchmark** of learning-augmented online-matching algorithms — advice-free baselines, the MPD family, and the test-and-fallback schemes — on one harness with a common error model, a common optimum, and confidence intervals (Section 3). - **(C2) The robustness-insurance characterization.** Unguarded followers crash below the advice-free floor; two mechanisms (structural and adaptive) restore safety with distinct trade-offs; the consistency upside is small on average-case inputs, so the value is downside protection. Validated on synthetic graphs, six real graphs, and real traces (Sections 3, 6). - **(C3) An empirical engagement with the order-error theory.** MPD's loss is governed by a Kendall-τ order error onto which several error models collapse; we characterize the tightness and saturation of the known n−LIS bound [ACI22, App. D] rather than re-deriving order-dependence (Section 4). - **(C4) The first empirical study of test-and-fallback**, including a threshold-calibration pathology (a more accurate test can make a worse decision), its recalibration and resolution limit, a **constant-free payoff-testing rule** that avoids both threshold pathologies — its misjudgement *falls* with prefix size exactly where the thresholds' *rises* — and a benchmark of the dynamic combiner exhibiting an irrevocability penalty that explains why matching needs *test-then-commit* (Section 5). - **(C5) The budget–stakes law.** A two-sided law for test-and-fallback — the two sides meeting up to logarithms under a stated signal condition (§7.5, §7.8): the follow/fallback decision costs a prefix of $\tilde{\Theta}(\sigma^2/g^2)$ (baseline slack over the square of the stakes) for *any* rule (lower bound), and an explicit directional test achieves it (upper bound). A payoff identity separates cheap payoff-testing from provably hard distance-testing — refuting, and reporting honestly, the natural tolerant-testing impossibility — and the corollary quantifies the wall: on strong-baseline instances, upsides below $\Theta(\sqrt{(1-\rho_{\text{base}})/n})$ are uncapturable at any prefix length (Section 7).

The experiments and the law are two views of one fact. The experiments discover a wall — predictions buy insurance, not performance; the law prices the wall — verifying advice costs the inverse square of its stakes, and average-case stakes cannot cover the bill. We adopt an AI-inference serving instantiation as a running application case study (Section 8) rather than a novelty claim, and we are careful throughout to credit the prior results our findings build on and to state the scope of each claim.

2 Setup

2.1 Model

We work in the **known-i.i.d.** model of online bipartite matching. There is a set R of n **offline resources** and a **type graph** on r **online types**, where type ℓ has a neighborhood $N(\ell) \subseteq R$ (the resources able to serve it). An **instance** is generated by drawing m arrivals i.i.d. from a distribution p over the r types; the i-th arrival reveals its type (hence its neighborhood) and must be matched, immediately and irrevocably, to a currently unmatched resource in its neighborhood, or left unmatched. The type graph and p are known to the algorithm; the realized arrival sequence is not. The classical setting is $r = n$ and $m = n$ (each offline vertex a distinct type); we also use *few-types* instances ($r \ll n$), where a prefix distribution-test is statistically meaningful.

We measure the **competitive ratio** $\rho(\mathrm{ALG}) = \mathbb{E}[|\mathrm{ALG}|]/\mathbb{E}[|\mathrm{OPT}|]$, where OPT is a maximum matching of the *realized* instance, computed exactly by Hopcroft–Karp [13]. All reported ratios are Monte-Carlo estimates with 95% confidence intervals (Section 2.5). The advice-free baseline we compare against throughout is KVV Ranking [16], whose ratio we denote ρ_{base} and call the **baseline strength**.

2.2 Algorithms

All the greedy-type algorithms share one primitive. Given a rank (a total order) on the resources, **GreedyWithPermutation** matches each arrival to its available neighbor of *minimum rank*. The advice-free baselines are three instances of this primitive together with the two stochastic-matching algorithms:

- **SimpleGreedy** — GreedyWithPermutation under the identity rank.
- **Ranking** [16] — GreedyWithPermutation under a uniformly random rank; the baseline.
- **Feldman** [11] and **Jaillet–Lu** [14] — the known-i.i.d. stochastic-matching algorithms (a max-flow blue/red decomposition, and a cap-3 LP realized by max-flow with per-type list distributions, respectively).

Degree predictions (object A). MinPredictedDegree (MPD) [1] is GreedyWithPermutation ranked by ascending *predicted degree* $\mu \in \mathbb{R}^R$: it matches to the available neighbor predicted to be rarest. With μ constant it reduces to Ranking; with μ the true realized degrees it is the consistency ceiling (MinDegree). Following ACI, we also form the **augmentations** Feldman(MPD) and JailletLu(MPD), which use the MPD rank only as the tie-break of the greedy fallback, so the worst-case-optimal base matching carries the load.

Type-histogram advice (object B) and test-and-fallback. The advice is a count vector $\hat{c} \in \mathbb{Z}^r_{\geq 0}$ over types (equivalently a distribution $q = \hat{c}/\|\hat{c}\|_1$). From $\hat{c}$ we build an *advice graph* ($\hat{c}_\ell$ copies of type ℓ), compute a maximum matching $\hat{M}$ on it, and record, per type, the resources $\hat{M}$ uses. **FollowPrediction** (blind Mimic) routes every arrival according to $\hat{M}$. **TestAndMatch** [6, 9] instead Mimics a sublinear prefix of k arrivals, estimates the ℓ_1 distance between the prefix type distribution and q, and — if the estimate is at most a threshold τ — continues to Mimic, else falls back to Ranking. The two variants differ only in an early-exit rule and in τ (Choo: $\tau = 2(\hat{n}/n - \beta)$; BEM: $\tau = 2(\hat{n}/n)(1-\beta)/(1+\beta)$), where β is the worst-case baseline ratio. Faithfulness note: the papers' ℓ_1

tester [15] has no deployable implementation and the authors themselves fall back to an empirical-ℓ_1 proof-of-concept; we do the same, and mark it as such.

The combiner (a benchmarked baseline, not a contribution). We port the blind-follow-with-switching combiner of Chłędowski et al. [8] — run FollowPrediction and Ranking as shadow experts and follow the leader with hysteresis — to contextualize test-and-fallback. We benchmark it; we do not claim it.

2.3 Prediction objects and the structured error model

To compare algorithms that consume *different* prediction objects (degree vectors μ vs type histograms $\hat{c}$), we corrupt each object with a per-family knob and report the families in parallel; this heterogeneity is intrinsic and is precisely why no unified benchmark existed.

For **degree predictions** we perturb the true degrees $\mu^\star$ with four structured error models, each returning both an ℓ_1 *magnitude* and a normalized **Kendall-τ order error** in $[0, 1]$: *random-flip* (replace a fraction of entries by uniform noise), *systematic-bias* (a monotone rescale — order-preserving, hence Kendall-$\tau \equiv 0$ by construction), *adversarial* (reflect a fraction of entries, the most order-damaging), and *distribution-drift* (blend toward degrees from an independently drawn graph of the same family). Because MPD depends on μ *only through the induced order*, the order/magnitude distinction is not cosmetic — it is the axis Section 4 is about.

For **type-histogram advice** we perturb the realized counts $c^\star$ toward a concentrated (spiky) random target by a fraction $\eta \in [0, 1]$ ($\eta = 0$: perfect advice, $\hat{c} = c^\star$; $\eta = 1$: advice unrelated to the truth), reporting the induced $\ell_1(p, q)$ as the advice-error axis for the test-and-fallback experiments.

2.4 Consistency and robustness

For a prediction-consuming algorithm we call ρ under *perfect* advice its **consistency** and its worst-case ρ over *adversarial* advice its **robustness**. A prediction algorithm is useful only if it is consistent *and* never falls (much) below ρ_{base}; the tension between the two, and its fundamental limits for the test-and-fallback class, are the subject of Sections 5 and 7.

2.5 Experimental methodology and reproducibility

The harness is a small dependency-light Python codebase (NumPy/SciPy/NetworkX). We draw all randomness from independent, reproducible streams via NumPy's spawn-based generators, and use **paired trials**: within a comparison, every algorithm and every error level reuses the same graph, arrival sequence, OPT, and tie-break seed, so differences are attributable to the prediction alone. Confidence intervals are 95% normal-approximation half-widths over trials. The Phase-2 portion reproduces a subset of Borodin, Karavasilis & Pankratov [5]; every figure and table in this paper is regenerated from a fixed seed by a single script, listed with its output in Appendix A. Most of those scripts are self-contained and run on a clean checkout; the real-graph and real-trace results of Sections 6 and 8 first require external data, whose provenance and retrieval are recorded in Appendix A.4.

One implementation detail is load-bearing for that claim. Three preprocessing steps — Feldman's and Jaillet–Lu's suggested matchings, and the serving advice b-matching — take a maximum flow and consume its *decomposition*, which, unlike the flow value, is not unique; the solver picks one by iterating node containers. We therefore label those networks' nodes with integers rather than strings, since Python randomises string hashes per process and string labels made the chosen decomposition, and every number derived from it, differ between runs of the same script.

3 The Unified Benchmark

This section places all three algorithm families on one harness and establishes the paper's organizing thesis. We report competitive ratios (mean ± 95% CI) as a function of prediction quality, in three panels chosen to span the regimes each family was designed for.

3.1 Design

The families consume different prediction objects, so each is driven by its own corruption knob and reported in a parallel panel (Section 2.3); the shared harness — graphs, OPT, CI methodology, and the advice-free floor — is what makes the panels comparable.

Instance format and notation. Every panel uses the instance format of Section 2.1: n offline resources, r online request types, and m arrivals drawn i.i.d. from the type distribution; throughout this section $m = n$, so requests and resources are balanced. The panels differ *only* in the type graph connecting requests to resources — that single change is what moves the input between the regimes the two prediction families were designed for. The three are chosen so that a *degree* predictor has strong signal, weak signal, and no role at all, respectively:

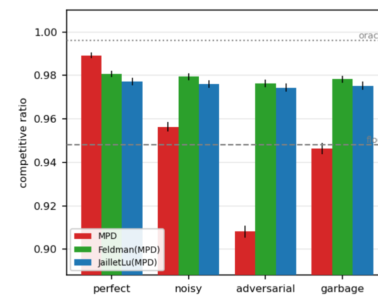
- **Panel A — heavy-tailed degrees (Zipf)** (n = 1000, 60 trials): resource degrees follow a Zipf power law with exponent 1.0 — a few resources are heavily contended while most are rarely eligible. This heavy-tailed profile gives a *degree* predictor genuine signal to carry.
- **Panel B — left-regular** d=5 (n = 1000, 60 trials): each arriving request connects to exactly $d = 5$ uniformly random resources, so resource degrees are nearly homogeneous — the hard case for this family, where a degree predictor has almost no signal left to carry.
- **Panel C — few-types** r=8 (n = 2000, 50 trials): only $r = 8$ distinct request types, each arriving $\approx n/r = 250$ times on average — the near-perfect-matchable, few-types regime the *histogram*-advice algorithms are calibrated for; their test-and-fallback test inspects a prefix of $k = 200$ arrivals.

Panels A and B thus exercise the degree-prediction family (MPD and its augmentations); Panel C exercises the histogram-advice family (FollowPrediction, TestAndMatch, and the combiner).

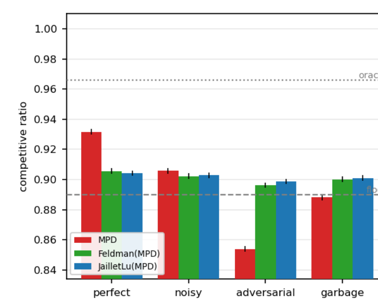
Shared methodology. Paired trials, independent random streams and confidence intervals are exactly as in Section 2.5; the panel-specific parameters are the ones listed above, and the intervals are tight throughout (±0.001–0.003). The quality columns instantiate the error models of Section 2.3 — degree panels: *perfect* (true realized degrees), *noisy* (random-flip at strength $\frac{1}{2}$), *adversarial* (order-reversing reflection), *garbage* (independent random μ, ≡ Ranking); advice panel: the true histogram blended toward a concentrated random target by $\eta \in \{0, 0.3, 0.6, 1.0\}$ (*perfect / mild / bad / garbage*). The two sets of columns are *not* commensurable — they corrupt different prediction objects with different knobs — so only within-panel comparisons carry meaning. **Table 1** presents each panel's ratios beside its bar chart; the findings follow in Section 3.2.

Table 1. The unified benchmark. Each panel's competitive ratios (means over paired trials; every 95% CI ≤ 0.003) sit beside their bar chart (error bars: 95% CIs; dashed line: advice-free floor; dotted: oracle ceiling). Bold marks the worst column of each unguarded prediction-follower; all three fall below their own panel's floor. That floor is instance-dependent — it is Ranking's ratio on that panel's graph family, not a constant — so the three floors (0.948, 0.890, 0.990) are not comparable with one another, and neither are the quality columns across the two prediction families (Section 2.3).

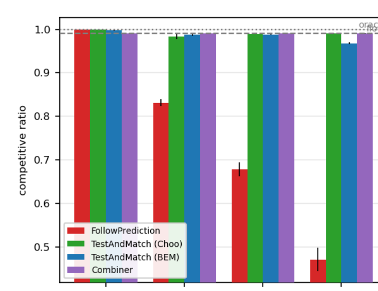
Panel A — heavy-tailed (Zipf)	perfect	noisy	advers.	garbage
Ranking (floor)	0.948	—	—	—
MinDegree (oracle)	0.996	—	—	—
MPD	0.989	0.956	**0.908**	0.946
Feldman(MPD)	0.981	0.979	0.976	0.978
JailletLu(MPD)	0.977	0.976	0.974	0.975
Feldman (base)	0.887	—	—	—
JailletLu (base)	0.900	—	—	—



Panel B — left-regular d=5	perfect	noisy	advers.	garbage
Greedy = Ranking (floor)	0.890	—	—	—
MinDegree (oracle)	0.966	—	—	—
MPD	0.932	0.906	**0.854**	0.888
Feldman(MPD)	0.906	0.902	0.896	0.900
JailletLu(MPD)	0.904	0.903	0.899	0.901
Feldman (base)	0.760	—	—	—
JailletLu (base)	0.788	—	—	—



Panel C — few-types r=8	perfect	mild	bad	garbage
Ranking (floor)	0.990	—	—	—
MinDegree (oracle)	0.999	—	—	—
FollowPrediction	1.000	0.832	0.679	**0.472**
TestAndMatch (Choo)	1.000	0.984	0.989	0.990
TestAndMatch (BEM)	0.998	0.988	0.988	0.968
Combiner *(benchmark)*	0.990	0.990	0.990	0.990



3.2 Four findings

(F1) Robustness is engineered, not free: naive followers crash below the floor. Both unguarded prediction-followers dive under the advice-free Ranking floor once the prediction is adversarial or garbage: MPD falls to $0.908 < 0.948$ (Panel A, adversarial) and $0.854 < 0.890$ (Panel B), and FollowPrediction collapses to $0.472 \ll 0.990$ (Panel C). Under adversarial or garbage advice, then, a practitioner using either *unguarded* is worse off than using no prediction at all — at perfect and noisy advice the same algorithms sit above the floor. Every *robust* algorithm in the tables — the augmentations, TestAndMatch, the combiner — avoids this by construction, not by luck.

(F2) Two distinct robustness mechanisms, with different shapes. *Structural* robustness (Feldman(MPD), JailletLu(MPD)): a worst-case-optimal base matching carries the load and the prediction only breaks ties, so performance is nearly *flat* across quality — Feldman(MPD) moves only $0.981 \to 0.976$ from perfect to adversarial (Panel A). It cannot crash, but it also caps the upside (it never reaches the 0.996 oracle, and sits below MPD's 0.989 at perfect). *Adaptive* robustness (TestAndMatch): test a sublinear prefix, then commit — capturing the upside when advice is good (Choo 1.000 at perfect) *and* holding the floor when advice is bad (0.990 at garbage). On Panel C it is the only algorithm on the upper envelope at both ends. The two mechanisms trade consistency for robustness in opposite ways.

(F3) The consistency upside is small on average-case inputs; the spread lives on the bad-advice side. On few-types the advice-free Ranking is already 0.990 and MPD-with-true-degrees is 0.999 — under 0.01 for *any* advice to add on the good side. Every wide gap in Panel C (1.000 → 0.472 for FollowPrediction; the half-wide envelope) is a *downside* gap. This is the thesis in one panel: learning-augmented matching on typical inputs is robustness insurance, not a performance lift — a fact Section 7 prices: safely capturing an upside this small would take a longer prefix than the instance itself.

(F4) The augmentation rescues structurally weak base algorithms. Feldman and Jaillet–Lu are tuned for the worst-case ratio and are the *weakest* advice-free entries on these average-case inputs (Panel B: 0.760 / 0.788, well below Greedy's 0.890). The MPD augmentation lifts them to ≈ 0.90 — the prediction does *more* for the worst-case-designed algorithms than for greedy. This pairing is visible only under a unified table.

3.3 Takeaway

Read together, F1–F4 say that on average-case matching the advice-free baseline is already near-optimal, unguarded prediction-following is unsafe, and the value of the sophisticated algorithms is downside protection delivered by one of two mechanisms. Sections 4–6 sharpen and stress-test each part — what governs the (small) loss, what the adaptive test costs, and whether the picture survives on real graphs, real traces, and a learned predictor — and Section 7 proves the wall is not an artifact but a price: a budget–stakes law that puts the required test prefix beyond the horizon exactly where the baseline is strong.

4 What Governs the Loss: Order-Error and ACI's Bound

MinPredictedDegree matches by ascending predicted degree, so it depends on the predictor μ *only through the order it induces on the resources.* Two predictors that induce the same order produce identical matchings; a monotone rescaling of μ changes nothing. The natural question is therefore not "how large is the prediction error" but "which *order* error governs the loss," and how tightly. This section answers that empirically. It also requires care: the qualitative fact that order — not magnitude — matters is *already theorem* in the literature, and we are explicit about what is prior and what is ours.

What is already known (ACI). Aamand, Chen & Indyk [ACI22, Appendix D] prove, for MinPredictedDegree on the CLV-B model, that the matching loss relative to the true expected degrees is at most $n - \mathrm{LIS}(w[\mu])$, where w is the vector of true CLV-B weights (unrelated to the type distribution p of Section 2.1), $w[\mu]$ is w ordered by the prediction μ, and LIS is the longest non-decreasing subsequence — a pure *order* quantity. In particular a monotone (order-preserving) predictor has $w[\mu]$ already sorted, so $\mathrm{LIS} = n$, $n - \mathrm{LIS} = 0$, and the loss is zero. Thus *order-dependence* and *the zero-effect of a monotone bias* are ACI's results, not ours; our `systematic_bias` error model (a monotone rescale) has Kendall-$\tau \equiv 0$ *by construction* and, consistently, leaves MPD's ratio exactly unchanged across the benchmark (Section 3) — an empirical confirmation of ACI's statement, not a new finding.

Our contribution is to characterize *how the bound behaves* and *which order measure predicts the realized loss.* We sweep the four structured error models across strength and record, per model and

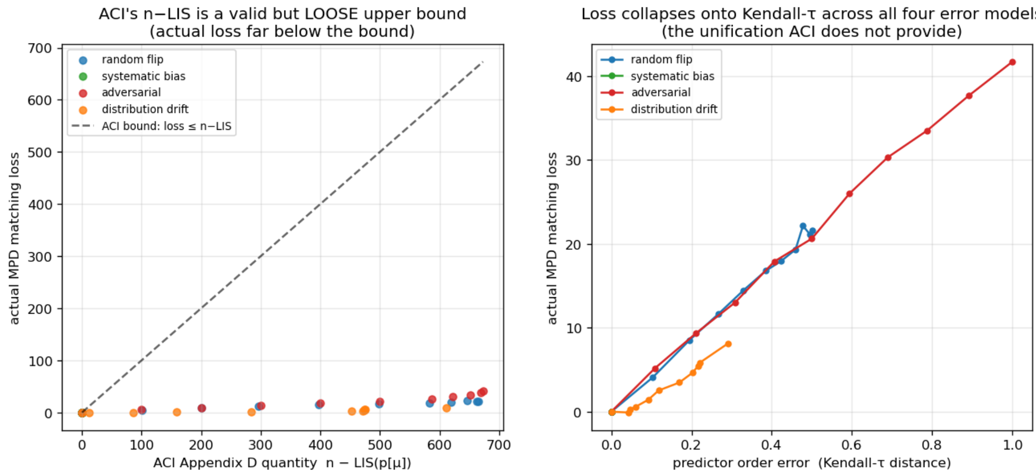


Fig. 1. Order error, not magnitude, governs MPD's loss: (a) realized loss sits far below the n − LIS bound at every point; (b) the four error models collapse onto the Kendall-τ axis.

level, three quantities on the same instances: the actual MPD matching loss, ACI's n − LIS, and the normalized Kendall-τ order error (**Figure 1**; $n = 1000$, Zipf exponent 1.0, 40 trials).

(i) ACI's n − LIS bound is correct but very loose. The realized loss sits far below the n − LIS upper bound at every point — by roughly 16× (adversarial) to 75× (distribution-drift). Figure 1(a) plots loss against n − LIS with the $y = x$ bound; all points lie close to the axis.

(ii) n − LIS saturates and cannot distinguish error structures. For every non-trivial error the quantity is pinned near its maximum (n − LIS ≈ 610–673 for $n = 1000$), even though the true losses differ by a factor of five across models — 8.1 (drift), 21.6 (random-flip), and 41.7 (adversarial) at full strength. As an upper bound that is (almost) always ≈ n, it carries little information about which prediction is actually more harmful.

(iii) Kendall-τ is the governing order measure, and the error models collapse onto it. Plotting the realized loss against the Kendall-τ order error (Figure 1(b)), the four structured error models fall on a single increasing curve: loss rises monotonically with Kendall-τ (0.29 → 0.50 → 1.0 for drift, random-flip, adversarial, tracking 8.1 → 21.6 → 41.7 in loss), with `systematic_bias` pinned at $\tau = 0$ and zero loss. Pooling all four models and all eleven corruption levels (44 points), the rank correlation between Kendall-τ and realized loss is $\rho_S = 0.979$ and the linear correlation is $r = 0.992$; within each individually non-degenerate model — the three whose τ actually varies — ρ_S is 0.973, 0.991 and 1.000. Collapse onto one curve is therefore a measured fact, not a visual impression. The quantity that *predicts* the loss — and unifies the error models — is the Kendall-τ order distance, which n − LIS (saturated) cannot resolve.

Takeaway (and scope). We do not claim that order, rather than magnitude, governs MPD — ACI's Appendix D establishes that, and the zero-effect of a monotone bias with it. What we add is a precise empirical characterization: ACI's n − LIS bound is loose by one to nearly two orders of magnitude and saturates, whereas Kendall-τ predicts the realized loss and unifies the four error structures onto one curve. This both sharpens the theory's practical content and justifies the single order-error axis we use when we stress-test the predictor on real data (Section 6).

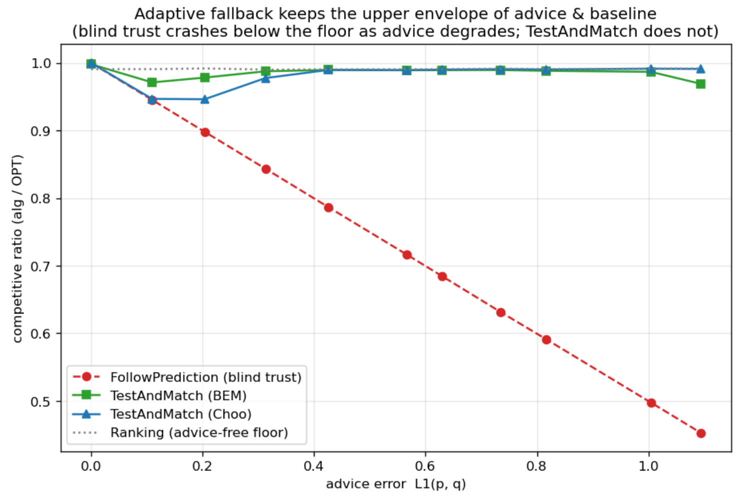


Fig. 2. The robustness envelope: blind FollowPrediction crashes below the advice-free floor as advice error grows; TestAndMatch stays on the upper envelope.

5 Test-and-Fallback in Depth

The test-and-fallback algorithms are the paper's adaptive robustness mechanism, and the object of the theory in Section 7. This section gives their first empirical study: the robustness envelope they achieve (§5.1), a counter-intuitive failure of the acceptance threshold (§5.2), its recalibration and the resolution limit that recalibration exposes (§5.3) — the empirical face of the budget–stakes law we prove in Section 7 — a **constant-free payoff-testing rule** that the law suggests and that avoids both threshold pathologies (§5.4), and the dynamic combiner benchmark that explains *commit-once* (§5.5). Throughout, the ℓ_1 test is the empirical-ℓ_1 proof-of-concept the original authors also fall back to (Section 2.2).

5.1 The robustness envelope

We sweep advice error $\ell_1(p,q)$ on few-types instances ($n = 2000$, $r = 8$, prefix $k = 200$, 40 trials) and compare blind FollowPrediction, TestAndMatch (Choo and BEM variants), and the Ranking floor (**Figure 2**). The picture is textbook. FollowPrediction degrades *linearly*, from 1.000 at perfect advice to 0.453 at $\ell_1 \approx 1.1$ — well *below* the advice-free floor (≈ 0.99). TestAndMatch instead stays on the **upper envelope**: it captures the benefit when advice is good and, when advice is bad, its prefix test rejects it and it falls back to Ranking. Across this sweep neither variant gives up more than 0.02 against the advice-free floor, versus FollowPrediction's 0.54 (BEM 0.998 $\longrightarrow$ 0.969; Choo 1.000 $\longrightarrow$ 0.991). That is a statement about this sweep — one graph family, 40 trials; the rule-independent version is Theorem 1. This is the adaptive counterpart of the structural robustness of Section 3: the engineered mechanism that MPD lacked.

5.2 A threshold that is too lenient on average-case inputs

What does the "test then maybe fall back" machinery cost, and does a better test make a better decision? We sweep the prefix (testing) size k at *borderline* advice ($\eta = 0.15$, true $\ell_1 \approx 0.16$) — the regime where the test must work hardest (**Figure 3**). Counter-intuitively, **a larger, more accurate test makes the *worse* decision**: as k grows 25 $\longrightarrow$ 100 $\longrightarrow$ 200 $\longrightarrow$ 800, the ratio

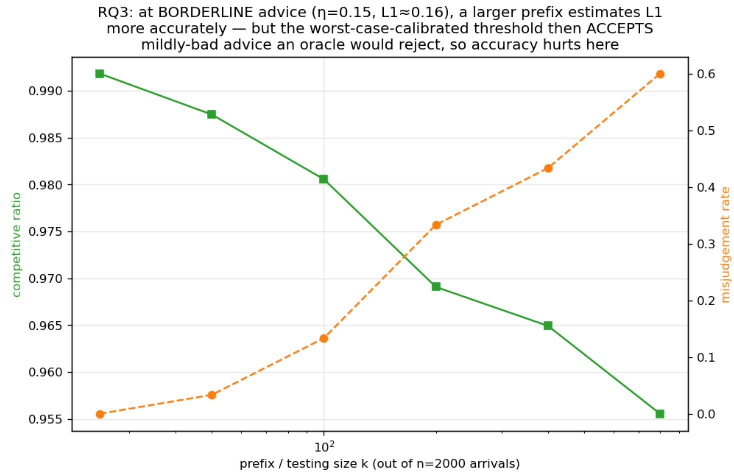


Fig. 3. Testing cost at borderline advice: under the worst-case-calibrated threshold, a larger and more accurate prefix test makes the *worse* decision.

falls 0.992 $\longrightarrow$ 0.981 $\longrightarrow$ 0.969 $\longrightarrow$ 0.956 and the misjudgement rate (against an empirical oracle: should we have followed, given that following actually underperformed the baseline here?) *rises* 0.00 $\longrightarrow$ 0.13 $\longrightarrow$ 0.33 $\longrightarrow$ 0.60.

The mechanism is a real and reportable finding. The Choo/BEM acceptance threshold τ is calibrated to the *worst-case* baseline $\beta \approx 0.696$ — the competitive ratio of Ranking under random arrival order [18], and the value Choo et al. plug into their threshold [9]; it is not $1 - 1/e$, which is the adversarial-order guarantee. But on these average-case instances the baseline is ≈ 0.99 (Section 3, F3), so the *empirical* break-even — the advice error at which following stops beating the baseline — sits at $\ell_1 \approx 0$, far below τ. A small, noisy prefix over-estimates ℓ_1 and *accidentally rejects* the borderline advice, landing on the strong Ranking floor; a large, accurate prefix correctly measures $\ell_1 \approx 0.16 < \tau$ and so **accepts** the mildly-bad advice the worst-case threshold deems acceptable — and underperforms the baseline. On strong-baseline inputs the worst-case-calibrated threshold is too *lenient*, and improving the test's accuracy only makes the algorithm follow that miscalibrated threshold more faithfully.

5.3 Recalibration, and the resolution limit it exposes

The natural fix is to recalibrate τ to the *measured* baseline $\hat{\beta}$ (the mean empirical Ranking ratio on a small calibration sample) rather than the worst-case β. On the same instances this eliminates the pathology: at borderline advice the worst-case threshold's misjudgement climbs 0.03 $\longrightarrow$ 1.00 as the prefix grows and its ratio drops to 0.920, whereas the recalibrated threshold holds misjudgement = 0.00 at every prefix size and ratio $\approx$ 0.986 — a larger prefix no longer hurts.

But recalibration exposes a deeper, honest limit. At *perfect* advice ($\ell_1 = 0$) the worst-case threshold scores 1.000 (it accepts and follows), while the recalibrated one scores only 0.987: the recalibrated threshold $\tau \approx 2(1 - \hat{\beta}) \approx 0.028$ is *smaller than the empirical-ℓ_1 estimator's own noise floor* ($\approx$ 0.05–0.13 at this prefix and support), so the test can never confidently *accept* — it rejects everything, including perfect advice, and effectively always plays the baseline. In other words:

> On strong-baseline instances, no practical empirical-ℓ_1 threshold can both capture the consistency upside and stay safe: the upside is tiny (the baseline is already $\approx$ OPT) and

sits *below the estimator's resolution.* The worst-case threshold over-accepts (§5.2); the recalibrated threshold over-rejects; and a better tester only follows whichever threshold more faithfully.

Section 7 turns this phenomenon into a two-sided law. The resolution limit above is specific to the empirical-ℓ_1 statistic — §7.7 shows that class is blind for a structural reason (the plug-in distance saturates at $k \ll r$ regardless of advice quality) — but the deeper point holds for *any* test on a sublinear prefix: the follow/fallback decision costs a prefix of $\tilde{\Theta}(\sigma^2/g^2)$ — baseline slack over the square of the stakes — and on strong-baseline instances like these the stakes are so small that the price exceeds the horizon. Section 5 is the empirical shadow; Section 7 is the law. But the resolution limit is a property of the *statistic*, not of testing itself — the next subsection builds the rule the law suggests, and it behaves differently.

5.4 Testing the payoff instead: a decision rule with no constants

The budget–stakes law identifies the decision-relevant statistic: not the advice's distance to the truth, but the *payoff* of following it. We implement this as **DirectionalTest-and-Match**: the same commit-once protocol (Mimic the prefix, decide once), with the ℓ_1 threshold replaced by a plug-in payoff comparison. From the prefix's empirical type distribution $\hat{p}$, the value of continuing to Mimic has an exact closed form, $V_{\text{mimic}}(\hat{p}) = \sum_l \min(n\hat{p}_l, s_l)$, where s_l is the number of offline slots the advice matching allocates to type l; the value of the Ranking baseline, $V_{\text{rank}}(\hat{p})$, is estimated by simulating Ranking on the *public* type graph under the same distribution. The rule follows iff the corrected difference is nonnegative. Two implementation findings are part of the paper's message. First, the naive plug-in is *biased*: V_{mimic} is concave, so honest sampling noise in $\hat{p}$ drags the estimate down at the capacity kinks (Jensen), and an uncorrected rule rejects even perfect advice; a parametric bootstrap anchored at $\hat{p}$ removes the bias (anchoring at the advice instead over-corrects exactly for mildly bad advice, whose true distribution sits in the locally linear region). Second, the rule uses **no problem constants**: no worst-case β, no threshold τ — every quantity is computed from the prefix and public information. A public-information early exit (fall back without spending the prefix when even a *correct* advice would not beat the baseline) restores parity with Choo et al.'s $\hat{n}/n \leq \beta$ exit.

Figure 4 repeats the envelope sweep with the payoff rule added, against both the worst-case and the recalibrated thresholds. The three rules now separate cleanly. At mildly bad advice — where the worst-case threshold follows the crash (0.948 and 0.937 at $\ell_1 \approx 0.10$ and 0.21, misjudging 100% and 63% of instances) — the payoff rule stays at the floor (0.989/0.990, misjudgement 3%/0%). At perfect advice — where the recalibrated threshold *never* follows (misjudgement 100%, §5.3) — the payoff rule captures the upside in ~40% of instances and lands safely on the floor otherwise. That partial capture is the honest ceiling, not a defect: at $k = 200$ the family's upside sits *at* the rule's resolution rather than above it. Rather than assume the variance, we measured it (`scripts/measure_payoff_variance.py`, Appendix A): drawing prefixes of length k under perfect advice, the decision statistic's standard deviation is 0.026/0.022/0.019/0.016/0.012 at $k = 50/100/200/400/800$; holding the prefix fixed and varying only the simulation seed accounts for under 4% of that variance at $k = 200$ and $k = 800$, so these are sampling numbers. Against this family's stakes ($g \approx 0.02$) that is a signal-to-noise ratio of 1.1 at $k = 200$ — precisely the regime in which a rule follows good advice sometimes and falls back otherwise. Nor does lengthening the prefix rescue it within the instance: a 16-fold longer prefix (50 $\rightarrow$ 800) buys only a 2.1-fold reduction in noise, leaving the margin at order one across the entire range we measured. We note one caveat in transferring Section 7 here: $k \cdot \text{Var}$ is not constant over that sweep (0.033 to 0.122), so the σ^2/k form proven on the cell family does not describe this estimator verbatim — the plug-in is

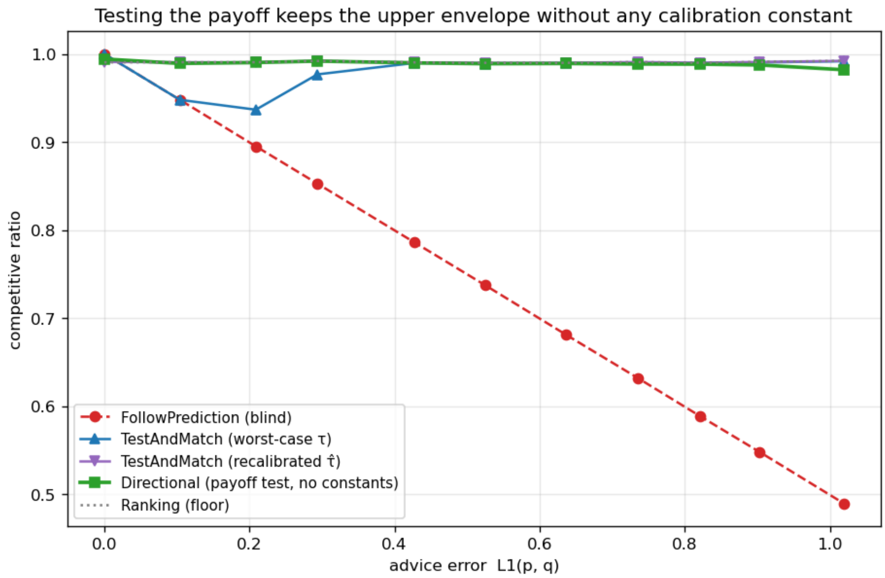


Fig. 4. The envelope sweep with the payoff rule added: the worst-case threshold follows the crash at mildly bad advice, the recalibrated threshold never captures at perfect advice, and the constant-free payoff rule tracks the upper envelope within the resolution the budget–stakes law allows.

a kinked functional of the prefix histogram, not a mean of per-sample observables — which is why we quote the measurement rather than the formula. The law's *qualitative* content binds our rule too, and the right behavior at barely-resolvable stakes is exactly this safe-either-way coin flip.

Figure 5 is the head-to-head that summarizes the section. At borderline advice ($\eta = 0.15$), growing the prefix from $k = 25$ to 800 drives the worst-case threshold's misjudgement *up* from 0.07 to 1.00 (ratio 0.987 $\rightarrow$ 0.922) — §5.2's pathology in full — while the payoff rule's misjudgement falls monotonically from 0.53 to 0.00 (ratio 0.954 $\rightarrow$ 0.991). More data helps if and only if the statistic being sharpened is the decision-relevant one; the recalibrated threshold is flat-safe here only because it always rejects. In one sentence: **test the payoff, not the prediction.**

The honest costs: when advice is bad but not worthless-if-true, the payoff rule pays the same prefix cost as its competitors ($\approx$ 1 point at $k/n = 0.1$; the early exit rarely fires on spiky advice), and its simulation budget — a handful of Ranking runs on the type graph — is computation, not samples. All numbers: 30 paired trials per point, seed-fixed (`scripts/run_directional_test.py`, `results/directional_test.json`).

5.5 The dynamic combiner is dominated, and shows why matching needs test-then-commit

Finally we benchmark the Chłędowski-style dynamic combiner (Section 2.2) to contextualize the *commit-once* structure of test-and-fallback. In its intended robust tuning the combiner sits exactly on the floor (0.990 across all advice quality on few-types): it never crashes, but captures *none* of the consistency upside — strictly dominated by TestAndMatch on the good-advice side and equal to it on the bad side.

More instructively, an *eager* combiner that switches experts mid-stream reveals a penalty specific to irrevocable problems. On a single-instance mechanism check ($n = 600$, $r = 6$, one seed, no

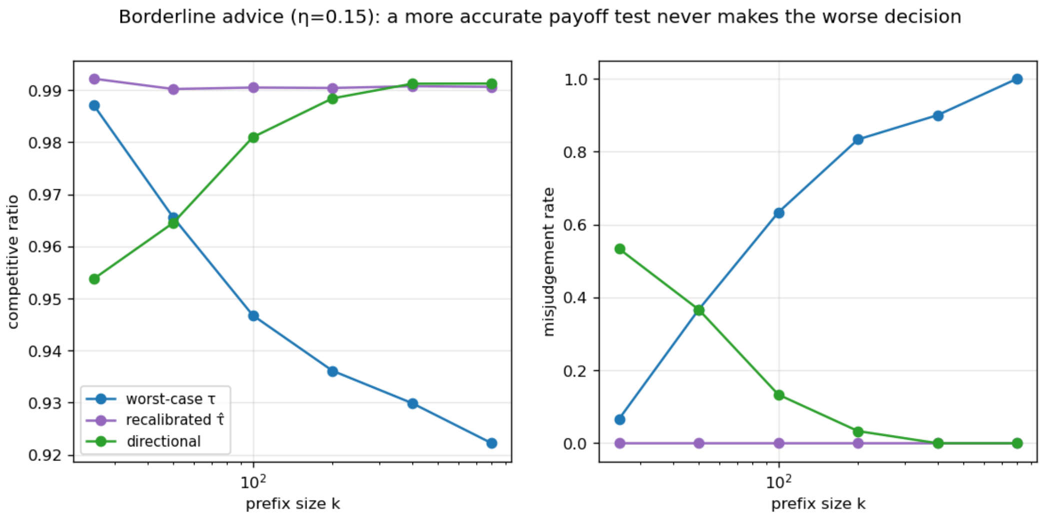


Fig. 5. Borderline advice (η = 0.15): as the prefix grows, the worst-case threshold's misjudgement rises to 1.0 while the payoff rule's falls to 0.0 — more data helps exactly when the statistic is the decision-relevant one.

averaging — a demonstration of the mechanism, not a measurement of its size), eager switching under perfect advice scores 0.927, *below both* the pure follower (1.000) and the pure baseline (0.958), because switching from Ranking to Mimic mid-run lands the committed matching in an *incompatible hybrid*: the Mimic plan's slots are already half-consumed by the Ranking phase, and the Ranking progress is abandoned. In an irrevocable problem the follow/fallback decision must be made *before* the bulk of the commitments, not adapted across them — which is precisely why Choo/BEM test on a prefix and then *commit*, rather than switching dynamically as the caching-style combiner does. The dynamic combiner that is cheap worst-case insurance for caching does not port cleanly to matching, and the unified benchmark shows why.

6 External Validity

Sections 3–5 are on synthetic graphs with a synthetic error knob. Is the picture real? We stress it three ways: a genuine, cheap predictor on a real request trace (§6.1); the six real-world graphs (§6.2); and whether *learning* the predictor changes anything (§6.3).

6.1 A real, cheap predictor

We replace the synthetic error knob with the cheapest realistic predictor: last-window historical statistics. Real Wikipedia daily pageviews give a live day (the truth) and earlier days (a 1-, 7-, or 30-day-stale forecast), so the prediction error is genuine temporal drift; we map the trace onto a fixed serving topology and consume the forecast through the degree route (MPD) (**Figure 6**; setup and per-staleness numbers in Appendix A).

Three facts emerge, all favorable and all honest. First, **the cost premise does not bite**: the predictor is a linear-time count ($\mu = A\,p$ over historical frequencies), 0.108 ms per instance — about 2% of the cost of computing OPT once — not an ML inference. Second, **the benefit is real, partial, and never harmful**: a stale forecast captures 27%–68% of the achievable oracle gap (falling with staleness) and *always* stays above the forecast-free baseline (0.938–0.957 vs 0.923–0.925, even at 30 days). Third, and explaining why, **topology aggregation makes the cheap predictor**

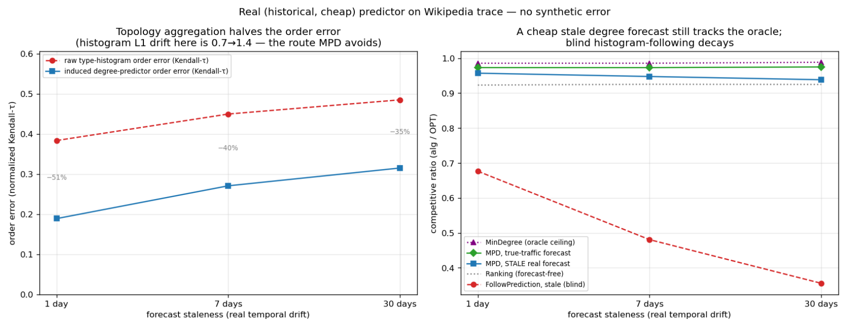


Fig. 6. A cheap historical-count predictor on the Wikipedia trace: near-oracle ratio at every staleness level, never below the advice-free floor.

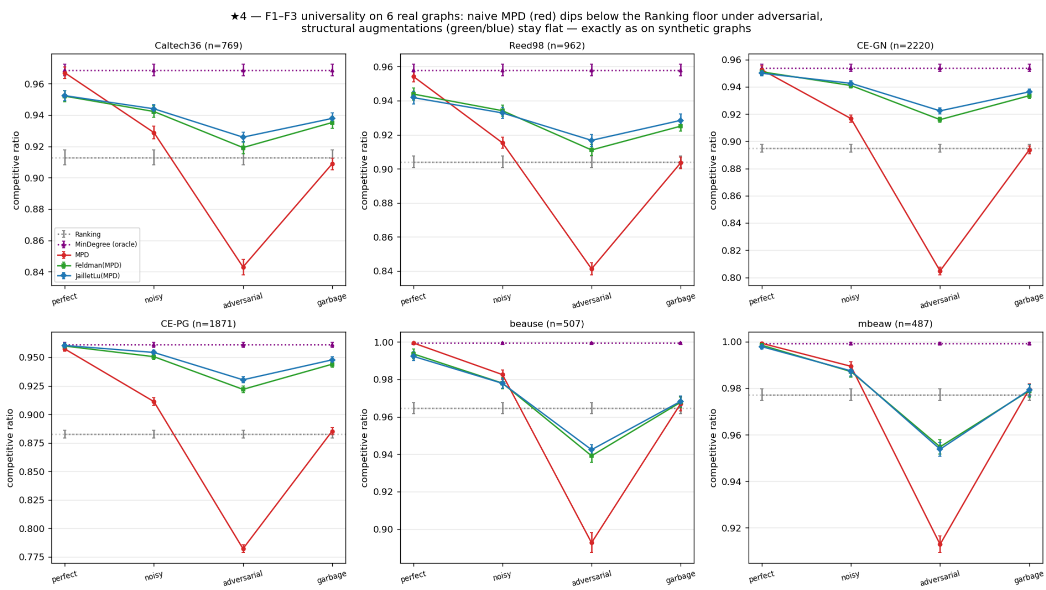


Fig. 7. The six real-world graphs: unguarded following falls below the advice-free floor on every graph, and the augmentation restores safety — the benchmark's findings survive on real topologies, not only on the generator.

order-faithful: the induced degree predictor's order error is only Kendall-$\tau \approx 0.19$–0.32, roughly *half* the raw type-histogram's (0.38–0.49) — and since MPD depends only on order (Section 4), the aggregated degree route survives real drift. Consuming the *same* stale forecast through the raw histogram instead is catastrophic: blind FollowPrediction collapses to $0.68 \longrightarrow 0.36$, far below the ≈ 0.92 baseline — exactly what the robust algorithms of Sections 3 and 5 are for.

6.2 Six real-world graphs

We re-run the degree-prediction roster of Section 3 on the six Network-Repository graphs (two Facebook social, two C. elegans biological, two economic input-output), across the same quality columns, with 95% CIs (**Figure 7**; graph sources and per-graph tables in Appendix A).

Both load-bearing findings hold on all six graphs. **F1**: naive MPD fed an adversarial predictor falls *below* the Ranking floor everywhere — by 0.06 (Reed98: $0.841 < 0.904$) to 0.10 (CE-PG: $0.782 < 0.883$). **F3 holds on all six, and confirms its own logic**: the consistency upside (best prediction algorithm at perfect advice, over the floor) is small everywhere (mean +0.049; range +0.022–+0.077), and it is *smallest exactly where the baseline is strongest* — the two dense economic graphs, with Ranking already 0.965/0.977, give the tiniest upsides (+0.035/+0.022).

The structural-robustness finding **F2 holds qualitatively on all six** (the augmentations are always less sensitive to prediction quality than naive MPD) and *strictly* on the four social/bio graphs (augmentation spread 0.22–0.29× MPD's); on the two dense economic graphs the protection is only *partial* — the augmentation cushions the adversarial drop (beause: 0.939 vs naive MPD's 0.893) but cannot fully clear the unusually high 0.965 floor, dipping ≈ 0.03 below it (spread ratio 0.53–0.55). This econ boundary is instructive rather than a failure: those graphs are so dense that matching is nearly trivial (Ranking $\approx$ 0.97, MinDegree = 1.00), so there is neither upside to capture (F3) nor much downside to protect — the mechanisms compress together where robustness does not matter. Finally, F4 is *dramatic* here: the worst-case-designed Feldman/Jaillet–Lu are the weakest advice-free entries on the econ graphs (0.73–0.77), and the MPD augmentation lifts them to 0.99 — a +0.26 rescue.

6.3 Does *learning* the predictor help? A negative result

Because MPD consumes the predictor only through its order (Section 4), one might train the predictor with an order-aware (rank) loss rather than regression. We test this and report an honest negative (training setup and the full configuration list in Appendix A). With *deliberately divergent* synthetic features — one carrying magnitude, one carrying order — rank-training does beat regression sharply (matching ratio 0.989 $\approx$ oracle vs 0.974, with rank-training having *worse* regression error but better order). But the advantage is gated by two things that both vanish in practice: it is zero when features do not induce an order/magnitude divergence and zero on easy instances (F3 again), peaking at only +1.3% on synthetic graphs; and, decisively, **on real temporal features it disappears** — trained on lagged counts from real serving traces, the rank- and regression-trained predictors produce *identical* order (Kendall-τ 0.126 vs 0.126) and identical matching ratio, across every topology and lag we tried. The dissociation that powers order-aware training is a property of engineered features, not of realistic ones. The lesson reinforces the thesis: once a predictor is order-faithful — which a cheap historical count already is (§6.1) — neither a better algorithm nor a better-trained predictor buys much on average-case matching.

6.4 Summary

On real predictors, real graphs, and a learned predictor, the same wall stands: the advice-free baseline is near-optimal, unguarded following is unsafe, and predictions buy downside protection rather than performance. Having established the wall empirically across every setting we could reach, we now prove it is necessary.

7 A Budget–Stakes Law: What a Sublinear Test Can and Cannot Buy

Every experiment above met the same wall: on average-case matching the advice-free baseline is near-optimal, so predictions buy robustness insurance rather than performance, and — for the test-and-fallback algorithms — no acceptance threshold both captures the (tiny) upside and stays safe (§5.3). This section quantifies the wall. The result is a two-sided **budget–stakes law**: on a natural instance family, deciding whether to follow the advice requires a prefix of length $k^* = \tilde{\Theta}(\sigma^2/g^2)$ — the instance's specialist mass (exactly twice its baseline slack) divided by the square of the stakes g — with **no** decision rule whatsoever simultaneously consistent and robust below the budget

(Theorem 1(i)) and an explicit, embarrassingly simple rule succeeding at it (Theorem 1(ii)). The two sides meet, up to logarithms, whenever the stakes are carried at comparable signal levels by a constant fraction of the specialist mass — the condition under which "k^*" names one quantity rather than two; when instead the stakes hide in a vanishing sliver of low-signal cells the sides can separate, and we leave that regime open (§7.8). The corollary is the wall: stakes are capped by the baseline slack, so on strong-baseline instances meeting that condition the required prefix exceeds the entire horizon — any upside smaller than $\approx 2\sqrt{(1-\rho_{\text{base}})/n}$ is uncapturable at *every* $k \le n$, and the upsides we measured in §3–§5 sit in exactly that range.

En route we record an honest negative of independent interest (§7.7): the natural conjecture — that the $\tilde{\Theta}(r/\log r)$ hardness of *tolerant distribution testing* [7, 15, 19] makes the follow/fallback decision itself hard for any rule at sublinear k — is **false**. The advice's *distance* $\ell_1(p,q)$ is indeed hard to test, and the empirical-ℓ_1 rules the field actually uses [6, 9] are provably blind at $k \ll r$; but the decision-relevant functional is the *payoff* of following, which on decomposable instances collapses to a per-sample observable. Testing the payoff is exponentially cheaper than testing the prediction.

Relation to prior work. Choo et al. [9] and Burathep–Erlebach–Moses [6] give *upper* bounds (algorithms) in this model and no statistical lower bound; Choo et al.'s Thm 3.1 is the generic adversarial trade-off (no algorithm is 1-consistent and $> \frac{1}{2}$-robust), driven by adversarial instances, not by the resolution of a sublinear test. Their acceptance threshold $\tau = 2(\hat{n}/n - \beta) - \varepsilon$ couples to the baseline β *constructively*; Theorem 1(i) is the information-theoretic counterpart — it binds *every* measurable rule on the prefix. The consistency/robustness lower bounds elsewhere in the with-predictions literature [21] are problem-intrinsic Pareto frontiers, not gated by a test. Tolerant-testing sample complexity [7] enters our story twice, both times off the critical path: it explains *why* the field's ℓ_1-threshold rules are blind (§7.7), and its hard instances are exactly what our payoff identity (Lemma 2) sidesteps. To our knowledge, both the budget–stakes law and the observation that payoff-testing strictly separates from distance-testing in an online algorithm are new. A dedicated pass over the adjacent lines (§9) found none testing the payoff: the *test-before-trust* program of Choo, Gouleakis and coauthors [4, 9] validates advice by distributional distance throughout — exactly the statistic §7.7 shows is the wrong one on average-case inputs; switching frameworks [2, 8] switch dynamically on prediction quality or regret rather than committing on a payoff test, and matching's irrevocability breaks them (§5.5); data-driven algorithm selection [3, 12] chooses among algorithms from samples of *whole instances* with observable per-instance performance, whereas here a single instance's prefix must reveal a policy's full-horizon value — which is what the payoff identity provides; and the $1/g^2$ budget echoes classical sequential analysis [20], whose engine — not its novelty — Lemma 2 imports into matching.

7.1 The test-and-fallback class

Fix an instance family and a budget $k = o(n)$. A **test-and-fallback** algorithm A_k observes the first k arrivals $X_{1:k}$ and the advice σ, outputs a (possibly randomized) decision $D \in \{\mathsf{Follow}, \mathsf{Fallback}\}$ as a function of $(X_{1:k}, \sigma)$, and thereafter Mimics σ if $D = \mathsf{Follow}$ and runs the advice-free baseline B (Ranking) otherwise. Because $k = o(n)$, the prefix's own contribution to the ratio is $O(k/n) = o(1)$, absorbed below. Write $\rho_{\text{base}} = \mathbb{E}[B]/\mathrm{OPT}$ for the baseline strength, **consistency** for the ratio under good advice, and **robustness** for the ratio under adversarial advice.

7.2 The master trade-off (rigorous)

Lemma 1. Let G, Bd be two instance distributions sharing the *same* advice σ, with prefix laws $\mathcal{L}_G, \mathcal{L}_{\mathrm{Bd}}$ and $\gamma_k := \mathrm{TV}(\mathcal{L}_G, \mathcal{L}_{\mathrm{Bd}})$. Suppose following σ gains δ under G

and loses Δ under Bd relative to B. Parameterize A_k by the fraction η_c of the upside it forgoes under G and its robustness loss $\eta_r\Delta$ under Bd. Then

$$(1-\eta_c) \;\le\; \eta_r + \gamma_k + o(1).$$

Proof. Conditioning on D under G, $\mathbb{E}_G[A_k]/\mathrm{OPT} = P_G(\mathsf{F})\,\mathbb{E}_G[\mathrm{Mimic}]/\mathrm{OPT} + P_G(\mathsf{R})\rho_{\mathrm{base}} \pm o(1) \ge \rho_{\mathrm{base}} + \delta\,P_G(\mathsf{F}) - o(1)$, so the captured upside $(1-\eta_c)\delta$ forces $P_G(\mathsf{F}) \ge 1-\eta_c-o(1)$. Symmetrically $P_{\mathrm{Bd}}(\mathsf{F}) = \eta_r + o(1)$. Since D is a function of $(X_{1:k},\sigma)$ and σ is identical, $|P_G(\mathsf{F}) - P_{\mathrm{Bd}}(\mathsf{F})| \le \gamma_k$; chaining gives the claim. □

With an *uninformative* prefix ($\gamma_k = o(1)$) one cannot be both near-fully-consistent ($\eta_c \to 0$) and robust ($\eta_r \to 0$). Everything below is a computation of how large k must be before the prefix becomes informative — and (the half the earlier draft missed) of how *small* a k already suffices.

7.3 The construction and the affine law

The family is a disjoint union of m independent **rare-resource cells**, each over a distinct type-pair, so the support is $r = \Theta(m)$ and (at $m = \Theta(n)$) each type is seen $O(1)$ times. Cell i has resources $\{a_i, b_i\}$, a flexible request (neighborhood $\{a_i, b_i\}$) that must be routed before a specialist ($\{a_i\}$ or $\{b_i\}$, present with probability θ_i) is seen; the right route depends on the future specialist. The profile $\{(\theta_i, \varepsilon_i)\}_{i\le m}$ is **arbitrary** — no homogeneity is assumed anywhere below. In closed form (verified numerically), cell i has $\mathrm{OPT} = 1+\theta_i$, baseline $1+\theta_i/2$, and following the advice gains or loses exactly $\theta_i|s_i - \frac{1}{2}|$ over the baseline according to whether the advice's *direction* is right, where $s_i = \frac{1}{2} \pm \varepsilon_i$ is the specialist bias and ε_i the signal; the advice-to-truth distance is $\ell_1 = 2\theta_i|s_i - \hat{s}_i|$ per cell.

These aggregate cleanly. Summing over cells yields an **exact affine law** (proven; verified to three decimals):

$$\mathbb{E}[\text{follow-ratio}] \;=\; \rho_{\mathrm{perfect}} - \tfrac{1}{2}\,\ell_1(p,q), \qquad \ell_1^\star \;=\; 2\big(\rho_{\mathrm{perfect}} - \rho_{\mathrm{base}}\big),$$

where ρ_{perfect} is the all-correct-direction ratio and $\ell_1^\star$ the break-even. The quantity that organizes everything is the family's **specialist mass**

$$\sigma^2 \;:=\; \frac{\sum_i \theta_i}{N}, \qquad N = \sum_i (1+\theta_i) = \mathrm{OPT}$$

— the probability that a random arrival is a specialist, and (below) the variance bound on the decision statistic. Two exact identities anchor what follows: the baseline slack *is* half the specialist mass,

$$1 - \rho_{\mathrm{base}} \;=\; \sigma^2/2,$$

and the maximum stakes (the full upside) are $\rho_{\mathrm{perfect}} - \rho_{\mathrm{base}} = \sum_i \theta_i\varepsilon_i/N \le 2\varepsilon_{\max}\,(1-\rho_{\mathrm{base}})$: *the upside is at most an ε-fraction of the baseline slack.* That the same σ^2 measures both the slack and the noise is not a coincidence — it is the law. The canonical (verified) instantiation takes $\theta_i = \theta$, $\varepsilon_i = \varepsilon$ and a wrong-direction fraction $\varphi = \frac{1}{4}$ (good: $\ell_1 = a$, gain δ) versus $\varphi = \frac{3}{4}$ (bad: $\ell_1 = b$, loss Δ), with $\delta = \Delta = \ell_1^\star/4$.

7.4 The payoff identity: the stakes are a per-sample observable

The earlier route tried to make the follow/fallback decision *hard* by making $\ell_1(p,q)$ hard to test. On this family that cannot work, for a reason worth a lemma. Classify each arrival by its role in its own cell, relative to the advice's predicted direction:

$$c(X) \;=\; \begin{cases} +1 & X \text{ is the advice-favored specialist of its cell,} \\ -1 & X \text{ is the disfavored specialist,} \\ 0 & X \text{ is flexible.} \end{cases}$$

Lemma 2 (payoff identity). For every type distribution p on the family — any profile $\{(\theta_i, \varepsilon_i)\}$, and any truth biases $s_i \in [0,1]$, matching the advice's magnitudes or not —

$$\mathbb{E}_p[c] \;=\; \frac{2}{N}\sum_i \theta_i\,(s_i - \tfrac{1}{2})\,\hat{d}_i \;=\; 2\cdot\frac{\mathbb{E}_p[\text{Mimic}] - \mathbb{E}_p[B]}{\text{OPT}}.$$

The expected advantage of following the advice is half the mean of a bounded, per-sample observable. (Verified to three decimals by `verify_witness_gap.py` for homogeneous wrong-fraction sweeps, and by `verify_budget_stakes_hetero.py` for random heterogeneous and magnitude-mismatched profiles.)

Proof. The advice-favored and disfavored specialists of cell i have masses $\theta_i\,(\frac{1}{2} \pm (s_i - \frac{1}{2})\hat{d}_i)/N$, where $\hat{d}_i$ is the advice direction; so cell i contributes $2\theta_i(s_i - \frac{1}{2})\hat{d}_i/N$ to $\mathbb{E}[c]$, while by the cell constants its follow-advantage is $\theta_i(s_i - \frac{1}{2})\hat{d}_i$. Sum over cells and divide by $\text{OPT} = N$. □

7.5 The budget–stakes law

Symbols. One quantity carries the stakes and one pair carries its two directions. For a scenario pair (G, Bd) sharing an advice, g is the **payoff gap** — the difference in follow-advantage between the two scenarios — while δ and Δ are the gain under G and the loss under Bd separately, the objects Lemma 1 is stated in. By definition $g = \delta + \Delta$, so $\min(\delta, \Delta) \le g/2$ always, with equality exactly when the pair is **balanced**. The flip pairs of (i) are balanced — flipping the cells of W moves each s_i symmetrically about $\frac{1}{2}$, giving $\delta = \Delta = g/2$ — so on them $\min(\delta, \Delta) = \Theta(g)$ and the two halves below constrain the same quantity. On an unbalanced pair (ii) is governed by the smaller of the two directions, and it is $\min(\delta, \Delta)$, not g, that sets the budget.

Theorem 1 (budget–stakes law, heterogeneous). On any cell family with profile $\{(\theta_i, \varepsilon_i)\}$ and specialist mass σ^2:

(i) (Impossibility below the budget — any rule.) Let G, Bd be the scenario pair that flips the advice-agreement of a cell set W whose signals are at most ε_W, with payoff gap $g = \frac{2}{N}\sum_{i\in W}\theta_i\varepsilon_i$. Every test-and-fallback algorithm A_k, deciding by *any* measurable rule on the prefix, with $k \;=\; o\big(1/(\varepsilon_W\, g)\big)$ has $\eta_c + \eta_r \ge 1 - o(1)$: it forgoes the upside or eats the loss.

(ii) (The budget suffices — an explicit rule.) For *any* pair of truth profiles — magnitudes need not match the advice — on which following gains $\ge \delta$ and loses $\ge \Delta$, the **directional test** — follow iff $\sum_{j\le k} c(X_j) \;>\; 0$ — achieves $\eta_c + \eta_r \;\le\; \epsilon_0$ once $k \ge C\,(\sigma^2/\min(\delta,\Delta)^2)\log(1/\epsilon_0)$. In particular k is *independent of n* for constant stakes. The two sides meet up to logarithms whenever the stakes are carried, at comparable signal levels, by a constant fraction of the specialist mass (Cauchy–Schwarz: $g^2 \;\le\; \frac{4}{N}\sigma_W^2\sum_W \theta_i\varepsilon_i^2$, σ_W^2 the flipped mass) — in particular for the canonical pair, giving the critical budget

$$k^* \;=\; \tilde{\Theta}\big(\sigma^2/g^2\big),$$

which at homogeneous parameters is the earlier $\tilde{\Theta}(\theta/\delta^2)$.

Proof of (ii). By Lemma 2, $\mathbb{E}[c] \;\ge\; 2\delta$ under the gain scenario and $\le\; -2\Delta$ under the loss scenario, for any truth profile; $\text{Var}(c) \;\le\; \mathbb{P}(\text{specialist}) \;=\; \sigma^2$; Bernstein's inequality gives error $\exp(-\Omega(k\min(\delta,\Delta)^2/\sigma^2))$ on both sides. □

Proof of (i), in outline (in full in Appendix B). Couple the two scenarios; the per-sample laws differ only on the specialists of the flipped cells, and the per-sample squared Hellinger distance is exactly $H^2 \;=\; \sum_{i\in W}\frac{2\theta_i}{N}\big(1 - \sqrt{1 - 4\varepsilon_i^2}\big) \;\in\; [4, 8]\cdot\sum_{i\in W}\theta_i\varepsilon_i^2/N \;\le\; 4\,\varepsilon_W\, g$. Tensorizing the Bhattacharyya coefficient and passing to total variation gives $\gamma_k \le \sqrt{k H^2} = o(1)$ whenever $k = o(1/(\varepsilon_W g))$, and Lemma 1 turns that into $\eta_c + \eta_r \ge 1 - o(1)$. □

Numerically (§`verify_witness_gap.py`, $\theta = 0.6$, $\varepsilon = 0.3$, $\delta = 0.056$): the directional test reaches $\eta_c + \eta_r \approx 0.06$ at $k = 100$ and 0.007 at $k = 200$, and stays flat as n grows from 3,200 to 320,000 at fixed $k = 200$ — the budget really is n-free once the stakes are constant. The heterogeneous claims are verified separately (`verify_budget_stakes_hetero.py`): the payoff identity holds for random profiles and for magnitude-mismatched truths; $1 - \rho_{\text{base}} = \sigma^2/2$ exactly; the per-sample Hellinger distance sits inside its $[4, 8]$ bounds; and three families with $\sigma^2 \in \{0.13, 0.24, 0.42\}$ produce decision-error curves that *coincide* as a function of kg^2/σ^2 alone (0.30/0.15/0.02/0.001 at $kg^2/\sigma^2 = \frac{1}{4}/1/4/8$) — the budget–stakes scaling is the whole story. Section 5.4 carries the rule from the hard family to the benchmark itself, where the plug-in payoff must additionally survive a concavity bias at the capacity kinks — it does, with a bootstrap calibration that still uses no problem constants.

7.6 The wall, quantified: strong baselines price the test out of the horizon

Stakes are capped by the slack: $g \le \rho_{\text{perfect}} - \rho_{\text{base}} \le 2\varepsilon_{\max}(1 - \rho_{\text{base}}) = \varepsilon_{\max}\,\sigma^2$. Theorem 1 places the feasibility frontier at $g \asymp \sigma/\sqrt{k}$: a prefix of length k resolves stakes down to $\sigma/\sqrt{k}$ and no further. With the whole instance as prefix ($k = n$), the constant-free threshold is

$$g^* \;=\; \sigma/\sqrt{n} \;=\; \sqrt{2(1 - \rho_{\text{base}})/n}\;:$$

> **Corollary (uncapturable upsides).** On strong-baseline families satisfying the matching condition of Theorem 1 — the stakes carried at comparable signal levels by a constant fraction of the specialist mass — every advice upside smaller than $\Theta\big(\sqrt{(1 - \rho_{\text{base}})/n}\,\big)$ is uncapturable by any test-and-fallback rule at any prefix length $k \le n$ — the information needed to decide does not exist inside the instance. (Theorem 1(ii) — stakes above the frontier *are* resolvable — is unconditional; it is the impossibility direction, and hence this corollary, that inherits the condition.)

At the parameters of our benchmark (§3: $\rho_{\text{base}} \approx 0.99$, $n = 2000$) the threshold is $g^* \approx 0.003$, and the measured upsides are < 0.01 (F3) — the empirical wall sits in the regime the corollary governs. This is the **scissors** of **Figure 8** (`results/impossibility_frontier.png`), now with the correct mechanism: the *potential* upside (perfect advice minus baseline) grows as the baseline weakens, while the upside a feasible test can safely capture is pinned by the $\sqrt{(1 - \rho_{\text{base}})/n}$ resolution — **the structure that makes the baseline strong is the structure that starves the test of signal.**

7.7 Why the field's rules hit the wall earlier: distance-testing is the wrong statistic

The algorithms of [6, 9] do not run the directional test; they threshold the plug-in distance $\hat{\ell}_1(\hat{p}_k, q)$. That class is blind far before Theorem 1(i) bites: at $k \ll r$ the empirical $\hat{p}_k$ is k atoms of mass $1/k$, so $\hat{\ell}_1 \approx 2$ *regardless of the advice's quality* (numerically: 1.908 vs 1.913 on the two canonical scenarios whose true ℓ_1 differ by 0.225). Any threshold therefore either always accepts or always rejects — the §5.3 pathology, now explained. The same blindness is measurable on the benchmark generator itself: holding the true error at $\ell_1 \approx 0.11$ and growing the support from $r = 8$ to 512 at fixed $k = 200$, the plug-in estimate inflates from 0.17 to 1.22 — twelve times the truth — while the payoff rule of §5.4 keeps deciding, staying within the prefix cost of the floor at both good and bad advice (**Figure 9**).

The deeper reason is the tolerant-testing barrier: estimating $\ell_1(p, q)$ to constant accuracy at $r = \Theta(n)$ genuinely requires $\tilde{\Theta}(n/\log n)$ samples [7]. The resolution of the apparent paradox — the *decision* is easy (Theorem 1(ii)) while the *distance* is hard — is Lemma 2: the payoff is a linear,

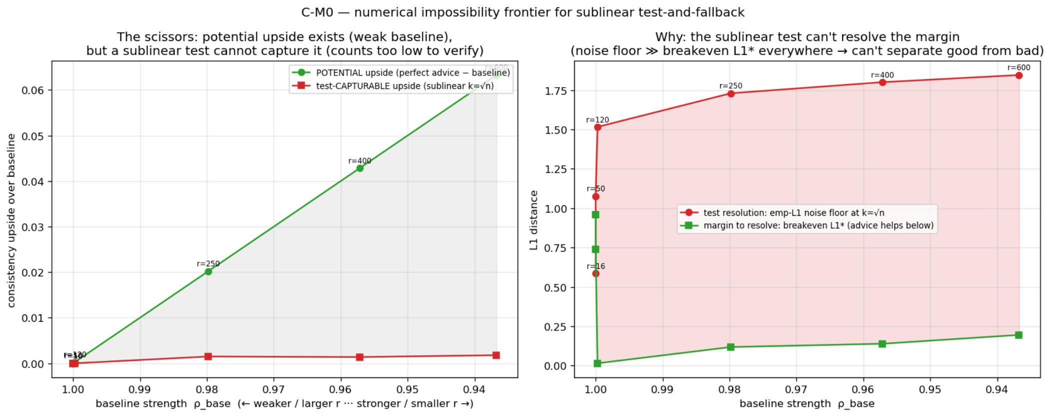


Fig. 8. The budget–stakes scissors: as the baseline weakens (left), the potential upside of perfect advice grows, while the upside a feasible test can safely capture stays pinned near zero — the empirical test's resolution sits far above the break-even margin wherever the upside exists.

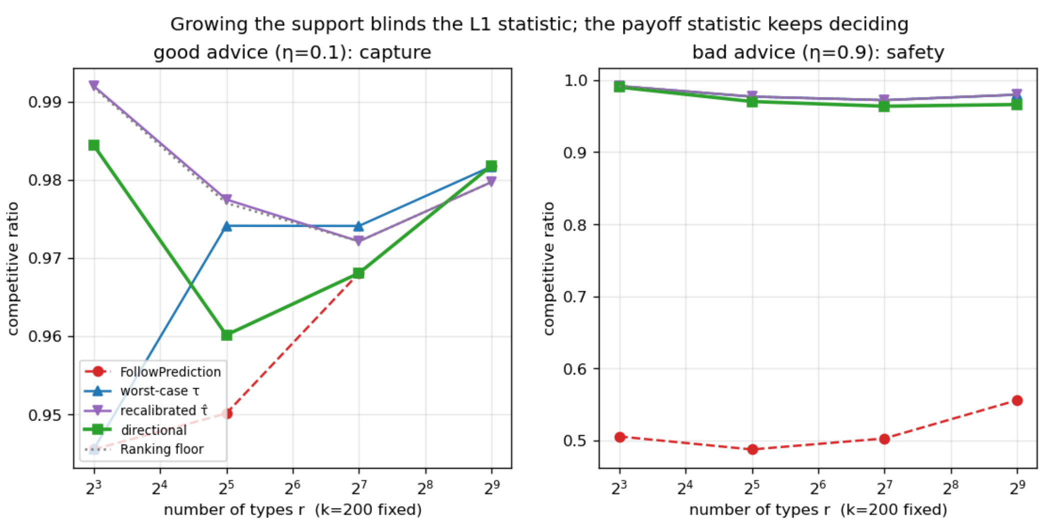


Fig. 9. The blindness curve on the benchmark generator: at fixed true $\ell_1 \approx 0.11$ and $k = 200$, the plug-in $\hat{\ell}_1$ inflates with the support size while the payoff rule keeps deciding at both good (left) and bad (right) advice.

per-sample functional, and ℓ_1 is not. The practical moral for algorithm designers is one sentence: **test the payoff, not the prediction.**

We record the flip side honestly. The natural conjecture behind an earlier version of this section — that tolerant-testing hardness makes the follow/fallback decision itself impossible for *any* rule at $k = o(n/\log n)$ with $\Theta(1)$ stakes — is **refuted** by Theorem 1(ii): no witness pair with $\gamma_k = o(1)$ and $\Theta(1)$ stakes exists on this family, because by Lemma 2 a $\Theta(1)$ stake gap forces $\gamma_k \longrightarrow 1$ at any $k = \omega(1)$. The tolerant-testing hard instances evade Lemma 2 only by mismatching the advice's per-cell magnitudes — precisely the regime in which distance and payoff decouple and ℓ_1 stops being the decision-relevant quantity.

7.8 Scope and honesty

Both halves of Theorem 1 use only elementary tools (Bernstein; Hellinger tensorization plus Lemma 1) — deliberately: the earlier, stronger-sounding route through tolerant-testing lower bounds is closed by §7.7, and we consider the refutation itself part of the contribution. The law now allows fully heterogeneous cell profiles, and its achievability side holds for arbitrary (magnitude-mismatched) truth distributions — neither homogeneity nor the matched-magnitude advice model is load-bearing. Its two sides can separate, by the factor $\sigma^2\varepsilon_W/g \geq 1$, when a scenario pair's stakes hide in an asymptotically vanishing sliver of low-signal cells; closing that regime is open. The law is stated for decomposable (disjoint-cell) families; by Lemma 2 the payoff is per-sample observable on *every* such family, so no decomposable construction can restore a super-$1/\delta^2$ barrier. Whether a *non-decomposable* family — long-range dependence making the payoff a genuinely hard functional — can separate the budget from $1/\delta^2$ is open. The directional test is analyzed here only on the cell family; a dedicated novelty pass — whose venues, keywords and cut-off are stated in §9 — found no payoff-testing acceptance rule in the adjacent literatures — the nearest lines test distances, switch on regret, or select algorithms from whole-instance samples (§9) — though the statistical skeleton of the upper half is, deliberately, classical. We credit Choo et al. for the constructive baseline-coupling their threshold already exhibits; our contribution is the two-sided, rule-independent budget law, the payoff/distance separation, and the quantified wall.

8 Application Case Study: AI-Inference Serving

To show the framework instantiates a contemporary problem, we cast request routing in an AI-inference serving system as online capacitated matching: resources are model replicas / cache shards serving up to c concurrent requests, arrivals are requests drawn from a (non-uniform, bursty) traffic distribution, and an edge is a capability or cache affinity. Two quantities must be kept apart: raw *goodput* is the fraction of arrivals served, while the *competitive ratio* is the number served divided by the capacitated optimum on the same realized instance; they coincide only when the optimum can serve every arrival, which under overload it cannot, so we report the competitive ratio. We instantiate the mapping across four serving concerns: capacity c (on a synthetic topology, where advice quality can be swept directly), stale traffic forecasts (Wikipedia pageviews), dynamic service times (an Azure LLM inference trace), and prefix-cache-aware routing (the Mooncake prefix-cache trace).

We present this as a **case study, not a novelty claim** — the systems facts we recover are established, and the "with-predictions" vocabulary is a re-labeling of them. The framework does reproduce them cleanly: capacity is a *substitute* for algorithmic robustness (blind trust in a forecast degrades further as capacity grows, while a capacity-aware baseline stays safe); a live-load signal beats a stale forecast under dynamic service times; and a *stable* cache-affinity router beats a reactive one — the reverse of the traffic-forecast case. Because the literature (§9) suggested the with-predictions lens might yield a *new* actionable serving result on a tail objective, we probed it: on an SLO/tail objective under bursty load, a non-predictive policy (static headroom or reactive-adaptive reservation) comes within $\leq 3\%$ of a policy with perfect foresight across every regime we swept. We found no regime where foresight helps — the tail objective is forgiving too, a third face of the paper's wall. Serving therefore stays a case study.

9 Related Work

Algorithm attributions are given where each algorithm is introduced (§2), and the positioning of our budget–stakes law against prior testing and tradeoff results is in §7; here we place the remaining landscape.

Scope of our novelty claims. Both claims we mark as new — the budget–stakes law, and the observation that testing the *payoff* separates from testing the prediction's distance — rest on a targeted search rather than an exhaustive one, and we state its extent so it can be checked. We searched the learning-augmented / algorithms-with-predictions literature through **April 2026**, over DBLP, arXiv (cs.DS, cs.LG) and Google Scholar, for prefix- or sample-based acceptance rules — keywords *test-before-trust*, *advice testing*, *trust the prediction*, *test-then-commit*, *prediction validation*, *consistency–robustness trade-off*, *tolerant testing* — and read in depth the four nearest lines (test-before-trust; switching and combining frameworks; distributional advice of unknown quality; data-driven algorithm selection), including the test-before-trust group's own April-2026 survey talk of their programme. We found no acceptance rule that estimates the payoff of following. A citation trace of every work citing [9] would be the completionist step and we have not performed it; the claims should be read as scoped to this search.

Learning-augmented algorithms. The consistency/robustness framework originates with competitive caching with predictions [17] and the optimal-tradeoff analyses that followed [21]; our thesis — that on average-case matching the value is robustness, not consistency — is a same-spirit but problem-specific statement, made quantitative and then proven necessary. The direct experimental-study template is Chłędowski et al. [8] in caching, whose blind-follow-with-switching combiner we benchmark (§5.5); the switching genre extends to prediction-quality-triggered frameworks such as SemiTrust-and-Switch [2], and distributional advice of unknown quality is handled by hedging in ski rental [10] and in Diakonikolas et al.'s sampling-access model — among the works covered by the search above, none tests the payoff and none commits once. The most closely related active line is the *test-before-trust* program of Choo, Gouleakis and coauthors [4, 9], which statistically validates advice before use — always by distributional distance; our §5.4/§7.7 argue the decision-relevant statistic is the payoff instead.

Algorithm selection and sequential testing. Choosing between two policies from samples at gap g costs $O(1/g^2)$ observations by uniform convergence — the classical engine of data-driven algorithm selection [3, 12] and, earlier, of sequential analysis [20]. The delta here is not that engine: in those settings each sample is a whole instance whose performance is directly observable, whereas a prefix of a *single* matching instance does not, in general, reveal a policy's full-horizon value. The payoff identity (Lemma 2) is the structural fact that makes it observable per-arrival on the hard family, and the bootstrap calibration of §5.4 is what survives of it on benchmark instances with capacity kinks.

Online matching with predictions. MinPredictedDegree [1] and the test-and-fallback schemes [6, 9] are the algorithms we unify and, for the latter, bound; each was previously studied in isolation. Borodin et al. [5] is the advice-free experimental foundation our benchmark extends.

Distribution testing. Tolerant identity testing [7, 19] and ℓ_1-distance estimation [15] enter our story as the explanation of why the field's empirical-ℓ_1 acceptance rules are blind at sublinear prefixes — the near-linear $\tilde{\Theta}(r/\log r)$ price of tolerance is real. Our budget–stakes law shows the follow/fallback *decision* nonetheless escapes that barrier: its decision-relevant statistic is the payoff, not the distance (§7.7), which is why our lower bound is a Hellinger computation rather than a testing reduction.

Serving systems. The systems results our §8 case study recovers are established across Preble, Mooncake, SageServe, and related work; we use them as ground truth, not as contributions.

10 Limitations and Conclusion

Limitations. (i) We work in the known-i.i.d. model; since Known-IID $\leq$ Random-Order, the algorithms' guarantees carry, but our empirical wall is an average-case statement and we do not claim it for adversarial arrival order. (ii) Following the original authors, the test-and-fallback experiments use an empirical-ℓ_1 surrogate for the (unimplemented) distribution tester; the theory does *not* inherit this modeling — the law binds any rule — and §7.7 separately explains the surrogate's blindness. (iii) The degree- and histogram-prediction families do not map onto every graph, which is why we report them in parallel panels rather than one table. (iv) Each real modality is exercised by one trace. (v) The budget–stakes law is proven on decomposable (disjoint-cell) families, where the payoff identity holds; whether a *non*-decomposable family can push the budget above $1/\delta^2$ is open. Our novelty claim for payoff-estimating acceptance rules rests on the search reported in §9: that search is complete, but bounded by the venues, keywords and cut-off stated there, so we present the claim as scoped rather than exhaustive. An earlier draft claimed a tolerant-testing impossibility for any sublinear rule; that claim was refuted during its own witness step and the refutation is reported in §7.7.

Conclusion. We gave the first unified experimental study of learning-augmented online bipartite matching — advice-free baselines, MinPredictedDegree and its augmentations, and the test-and-fallback algorithms — on one harness, with confidence intervals, across synthetic graphs, six real graphs, and real traces. Across every setting the same wall appears: the advice-free baseline is already near-optimal, unguarded prediction-following crashes below it, and the value of the sophisticated algorithms is downside protection, delivered by structural or adaptive robustness with distinct trade-offs. We then priced the wall for the test-and-fallback class with a two-sided budget–stakes law: the follow/fallback decision costs a prefix of $\tilde{\Theta}(\sigma^2/g^2)$ — baseline slack over squared stakes — for *any* decision rule, an explicit directional test achieves it (implemented and benchmarked, it avoids both threshold pathologies, §5.4), and — because stakes are capped by the baseline slack — on strong-baseline instances the price exceeds the horizon: upsides below $\Theta(\sqrt{(1-\rho_{\text{base}})/n})$ are uncapturable at any feasible prefix. Experiments discover the wall; theory sends the bill. The practical message is a single sentence — **on average-case matching, the advice's upside is smaller than the price of finding out whether to trust it** — with one constructive corollary for algorithm designers: *test the payoff, not the prediction* (the payoff is a per-sample observable where the distance is not). Progress on predictions for online matching will come from the regimes this paper brackets: adversarial or non-stationary arrivals, and objectives (tail, fairness, cost) where the advice-free baseline is genuinely far from optimal.

A Reproduction

Every number in this paper is regenerated from a fixed seed by a single script. Two classes must be distinguished. Most scripts are **self-contained**: they generate their own synthetic instances and run as-is on a clean checkout. The rest — the real-graph and real-trace results of Sections 6 and 8 — first need external data placed under `data/`, which we do not redistribute; §A.4 records what each dataset is and where it comes from. A dagger (†) marks that second class in the map below.

A.1 Environment and conventions

- **Stack:** Python 3.12; NumPy 1.26, SciPy 1.13, NetworkX 3.3, Matplotlib (plots only).
- **Randomness:** every stream derives from one master seed (default `0`), spawned into independent reproducible sub-streams with NumPy's `default_rng(seed).spawn(k)` — one each for the graph, the arrival sequence, algorithm randomness and the perturbation.
- **Paired trials:** within a comparison every algorithm and every error level reuses the same graph, arrival sequence, OPT and tie-break seed.
- **Confidence intervals:** 95% normal-approximation half-widths over trials.

- **Flow decompositions.** Feldman, Jaillet–Lu and the serving advice b-matching consume the *decomposition* of a maximum flow, which — unlike the flow value — is not unique. Their networks label nodes with integers rather than strings, so the decomposition NetworkX returns is stable across runs; under string labels it followed the interpreter's per-process string hashing and every derived number moved between runs of the same script (Section 2.5).
- Runtimes below are wall-clock on one machine (Apple M4 Pro, 12 cores). The scripts are single-threaded apart from NumPy's own vectorization, so they scale with single-core speed.

A.2 Result → script map

Each row names a *stem*. Unless the row says otherwise, its script is run_<stem>.py under scripts/, and its outputs are <stem>.json and <stem>.png under results/.

Paper object	Script stem (or path)	≈ time
Table 1 (unified benchmark)	unified_benchmark, then scripts/plot_unified_panels.py for the panel charts and _tables.md	100 s
Fig. 1 (order error vs the ACI bound)	order_vs_theory	30 s
§4 collapse statistics (ρ_S, r)	derived from order_vs_theory.json: rank and linear correlation of kendall against loss, pooled over all four models and all eleven levels	—
Figs. 2–3 (envelope; prefix pathology)	choo_bem	20 min
§5.3 recalibration	recalibration	1.5 min
Figs. 4, 5, 9 (payoff rule; r-sweep)	directional_test	12 min
§5.4 decision-statistic variance	measure_payoff_variance.py (not a run_ script)	30 min
§5.5 eager-combiner mechanism check	tests/test_combiner_small.py (console)	2 s
Fig. 6 (real predictor) †	real_predictor	15 s
Fig. 7 (six real graphs) †	realworld_robustness, plus its _tables.md	65 s
§6.3 rank vs regression training	rank_vs_mse_mve, rank_when_it_matters, rank_real_trace †	40 s
Fig. 8 (budget–stakes scissors)	impossibility_frontier	6 s
§7 numerical verification	scripts/verify_witness_gap.py and verify_budget_stakes_hetero.py (console)	seconds

Paper object	Script stem (or path)	≈ time
§8 serving case study †	serving, serving_trace, serving_dynamic, prefix_cache, serving_slo_probe	varies
Phase-2 reproduction of [5]	er_full, left_regular, realworld †	35 min

Scripts live under scripts/ and write to results/; all are run from the project root. Seven hand-verifiable correctness anchors live under tests/ (for t in tests/test_*.py; do python3 "$t"; done).

A.3 Order of execution

Two dependencies are not self-evident: plot_unified_panels.py and run_consistency_robustness.py both consume results/unified_benchmark.json and must run after run_unified_benchmark.py. Everything else is independent.

```
# self-contained (clean checkout, no external data)
python3 scripts/run_unified_benchmark.py       # Table 1 (data)
python3 scripts/plot_unified_panels.py      # Table 1 (panel charts; needs the line above)
python3 scripts/run_order_vs_theory.py         # Fig 1
python3 scripts/run_directional_test.py        # Figs 4, 5, 9
python3 scripts/run_impossibility_frontier.py  # Fig 8
python3 scripts/verify_witness_gap.py       # Lemma 2, directional test, plug-in blindness
python3 scripts/verify_budget_stakes_hetero.py # heterogeneous profiles, slack identity
python3 scripts/measure_payoff_variance.py     # sigma^2 quoted in 5.4
python3 scripts/run_choo_bem.py                # Figs 2, 3   (~20 min)

# require data/ (see A.4)
python3 scripts/run_real_predictor.py          # Fig 6
python3 scripts/run_realworld_robustness.py    # Fig 7
python3 scripts/run_rank_real_trace.py         # 6.3, real-feature arm
```

A.4 Data provenance

External data lives under data/ and is not redistributed with the code.

- **Real graphs (Section 6.2).** Six graphs from the Network Repository (networkrepository.com): socfb-Caltech36, socfb-Reed98, bio-CE-GN, bio-CE-PG, econ-beause and econ-mbeaflw, downloaded as MatrixMarket or edge-list files, reduced to simple undirected graphs, and made bipartite by a random balanced partition.
- **Traces (Sections 6.1, 6.3, 8).** Wikipedia "top articles per day" for four days from the Wikimedia REST pageviews API (data/trace/wiki/; the live day is the truth, earlier days are the 1-, 7- and 30-day-stale forecasts); the Azure LLM inference trace from Microsoft's public Azure dataset release (data/trace/azure_llm/; context and generated token counts with timestamps); and the Mooncake conversation trace (data/trace/mooncake/; per-request prefix-cache block hashes). The Wikipedia data is a snapshot of a live source rather than a static benchmark, so exact reproduction requires the same snapshot, not merely the same API.

B Proof of Theorem 1(i)

Section 7.5 gives the argument in outline; we record it in full here, since it is the lower half of the paper's main theorem. Recall the setting: a cell family with profile $\{(\theta_i, \varepsilon_i)\}_{i\le m}$, specialist mass $\sigma^2 = \sum_i \theta_i/N$ with $N = \sum_i(1+\theta_i) = \mathrm{OPT}$, and a scenario pair (G, Bd) that shares one advice and flips the advice-agreement of a cell set W whose per-cell signals satisfy $\varepsilon_i \le \varepsilon_W$. Its payoff gap is $g = \frac{2}{N}\sum_{i\in W}\theta_i\varepsilon_i$.

Claim. Every test-and-fallback algorithm A_k that decides by an arbitrary measurable (possibly randomized) function of the prefix $X_{1:k}$ and the advice, with $k = o\big(1/(\varepsilon_W g)\big)$, has $\eta_c + \eta_r \ge 1 - o(1)$.

Proof. By Lemma 1 it suffices to show $\gamma_k := \mathrm{TV}(\mathcal{L}_G, \mathcal{L}_{\mathrm{Bd}}) = o(1)$, where $\mathcal{L}_\cdot$ is the law of the prefix under each scenario; Lemma 1 then gives $(1-\eta_c) \le \eta_r + \gamma_k + o(1)$, i.e. $\eta_c + \eta_r \ge 1 - o(1)$.

Step 1 (the coupling). Arrivals are i.i.d., so $\mathcal{L}_G = P^{\otimes k}$ and $\mathcal{L}_{\mathrm{Bd}} = Q^{\otimes k}$ for the two per-arrival laws P, Q on the type set. The two scenarios differ only in the specialist bias s_i of the cells in W: outside W, and on every flexible type, P and Q assign identical mass, and the coupling that leaves those arrivals untouched is exact there. Inside a flipped cell $i \in W$, the two specialist types carry masses $\frac{\theta_i}{N}(\frac{1}{2}+\varepsilon_i)$ and $\frac{\theta_i}{N}(\frac{1}{2}-\varepsilon_i)$ under P, and the same two masses swapped under Q.

Step 2 (per-sample Hellinger distance). Writing $H^2(P,Q) = \sum_x(\sqrt{P(x)} - \sqrt{Q(x)})^2$, only the $2|W|$ swapped atoms contribute, and cell i contributes

$$2\,\frac{\theta_i}{N}\Big(\sqrt{\tfrac{1}{2}+\varepsilon_i} - \sqrt{\tfrac{1}{2}-\varepsilon_i}\Big)^2 \;=\; \frac{2\theta_i}{N}\Big(1 - \sqrt{1-4\varepsilon_i^2}\Big),$$

using $(\sqrt{a}-\sqrt{b})^2 = a+b-2\sqrt{ab}$ with $a+b=1$ and $ab = \frac{1}{4}-\varepsilon_i^2$. Since $2t \le 1-\sqrt{1-4t} \le 4t$ for $t = \varepsilon_i^2 \in [0, \frac{1}{4}]$,

$$H^2(P,Q) \;=\; \sum_{i\in W}\frac{2\theta_i}{N}\Big(1-\sqrt{1-4\varepsilon_i^2}\Big) \;\in\; [4,8]\cdot\frac{1}{N}\sum_{i\in W}\theta_i\varepsilon_i^2.$$

Bounding each ε_i by ε_W in one factor,

$$H^2(P,Q) \;\le\; \frac{8}{N}\sum_{i\in W}\theta_i\varepsilon_i^2 \;\le\; 4\varepsilon_W\cdot\frac{2}{N}\sum_{i\in W}\theta_i\varepsilon_i \;=\; 4\,\varepsilon_W\, g.$$

Step 3 (tensorization). With $h^2 := H^2/2 = 1 - \mathrm{BC}(P,Q)$ the normalized squared Hellinger distance, the Bhattacharyya coefficient of a product measure is the product of the coefficients, so $1 - h^2(P^{\otimes k}, Q^{\otimes k}) = (1-h^2(P,Q))^k$ and hence $h^2(P^{\otimes k}, Q^{\otimes k}) \le k\,h^2(P,Q)$.

Step 4 (to total variation). The standard comparison $\mathrm{TV} \le h\sqrt{2-h^2} \le \sqrt{2}\,h$ gives

$$\gamma_k \;\le\; \sqrt{2}\,h(P^{\otimes k}, Q^{\otimes k}) \;\le\; \sqrt{2k\,h^2(P,Q)} \;=\; \sqrt{k\,H^2(P,Q)} \;\le\; \sqrt{4k\,\varepsilon_W\, g}.$$

Step 5 (conclusion). If $k = o\big(1/(\varepsilon_W g)\big)$ then $k\,\varepsilon_W g \to 0$, so $\gamma_k \to 0$: the prefix is asymptotically uninformative about which scenario is in force, and Lemma 1 converts that into $\eta_c + \eta_r \ge 1 - o(1)$. Here $o(1)$ is taken along any sequence of families and prefix lengths with $k\varepsilon_W g \to 0$; the $o(1)$ inside Lemma 1 additionally absorbs the $O(k/n)$ contribution of the prefix itself to the ratio, which is why the sublinearity assumption $k = o(n)$ is in force. □

Where the two halves meet. Theorem 1(ii) spends $k = O(\sigma^2/\min(\delta,\Delta)^2)$, while (i) forbids $k = o(1/(\varepsilon_W g))$. On a balanced pair $\min(\delta,\Delta) = g/2$, so the gap between the two is the ratio $\sigma^2\varepsilon_W/g$. Cauchy–Schwarz on the definition of g gives $g^2 \le \frac{4}{N}\,\sigma_W^2\sum_{i\in W}\theta_i\varepsilon_i^2$ with σ_W^2 the flipped mass, so that ratio is $\Theta(1)$ — and the two sides agree up to logarithms — exactly when the stakes are carried, at comparable signal levels, by a constant fraction of the specialist mass. When instead they hide in a vanishing sliver of low-signal cells the ratio can diverge; that regime is open (Section 7.8).

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